

Announcements

There is now a "feedback" section on the SEMT30003 blackboard page:

- ▶ Predict and control exam marks via
 - Past exams for practice
 - Last year's exam + supplemental worksheets for readiness testing
 - Bring questions to instructors in exam review session to close gaps
- ▶ Predict and control coursework marks by
 - Asking demonstrators how labs relate to the extended coursework (they have advance access to a draft, although this may change)
 - ***Critical!*** Preparing a complete draft coursework early and bringing it to coursework support session for comments
- ▶ Email instructors to close gaps in the above (office hours for extended questions are possible)

The Week 2 in-lecture derivation of the LIF spiking period had a sign error.

- ▶ This is fixed in the uploaded annotations

Now on Github

- ▶ Annotated week 1, 2 slides
- ▶ Draft week 3, 4 slides

end

... Quantum Chemistry, Molecular dynamics

Molecules

Gillespie, Master Equation

Concentrations

Mass-Action Kinetics

Conductance Models

Hodgkin-Huxley

Spiking Models

Leaky Integrate and Fire

Rate Neurons

Neural Mass/Field Models

Poisson Neurons

Generalized Linear Models

Binary Neurons

McCulloch-Pitts, Hopfield, Perceptron

Cognitive Neuroscience, Psychology ...

Physiological,
Quantitative

Biological
Realism,
Data needed
to identify
parameters



Computational
Efficiency,
Mathematical
Tractability



Phenomenological,
Qualitative